



Bucharest University of Economic Studies
Faculty of Cybernetics, Statistics and Economic Informatics
Course: Time Series Analysis



EXAM PROBLEMS – PRACTICE SET

The problems have the form of exam problems: the interpretation of a software output, a short derivation or the choice of a model. The data are the series used in the course (last day: 18 September 2026), and the software outputs (statsmodels, arch, scikit-learn) are shown as the programs print them. The numerical answers are at the end of the document. Notation: ε_t white noise with variance σ^2 ; L the lag operator; $r_t = 100 \ln(P_t/P_{t-1})$ the log return in %; $\text{VaR}_\alpha = -q_\alpha$ (VaR 1% is the loss exceeded on 1% of the days).

Useful values: $z_{0.975} = 1.960$, $z_{0.95} = 1.645$, $\chi^2_{0.95}(1) = 3.84$, $\chi^2_{0.95}(2) = 5.99$, $\chi^2_{0.95}(3) = 7.81$, $\chi^2_{0.95}(4) = 9.49$, $\chi^2_{0.95}(5) = 11.07$, $\chi^2_{0.95}(6) = 12.59$, $\chi^2_{0.95}(8) = 15.51$, $\chi^2_{0.95}(10) = 18.31$, $\chi^2_{0.95}(12) = 21.03$, $\chi^2_{0.95}(20) = 31.41$; 5% Dickey–Fuller critical values: -1.95 (no constant), -2.86 (constant), -3.41 (constant and trend); $\ln 2 = 0.693$ (z_p and χ^2_p are quantiles of order p).

Chapter 1. Stochastic processes and stationarity

Problem 1 • The correlogram of the Nile flow

The annual flow of the Nile at Aswan (10^8 m^3 , 1871–1970, $n = 100$, mean 919.4) has the correlogram below.

Software output

Correlogram of the Nile annual flow, 1871–1970

Sample size: 100

Lag	AC	PAC	Q-Stat	Prob
1	0.498	0.498	25.59	0.000
2	0.385	0.181	40.99	0.000
3	0.328	0.111	52.29	0.000
4	0.239	0.006	58.37	0.000
5	0.228	0.065	63.97	0.000
6	0.227	0.071	69.58	0.000
7	0.222	0.060	74.99	0.000
8	0.300	0.163	84.96	0.000
9	0.142	-0.148	87.21	0.000
10	0.090	-0.065	88.13	0.000

- Compute the limits of the 95% confidence band for the sample autocorrelations under the white-noise hypothesis.
- State the decision of the Ljung–Box test with 10 lags at the 5% significance level.
- Identify the ARMA model suggested by the pattern of the ACF and the PACF.

Problem 2 • The US 10-year Treasury yield: level and change

Figure 1 shows the 10-year US Treasury yield (daily, 2015–2026) and its daily change in basis points. For the level, the first five sample autocorrelations are 0.998; 0.997; 0.995; 0.993; 0.992. For the daily change ($n = 2926$) they are -0.037 ; -0.036 ; 0.006 ; -0.012 ; -0.008 .

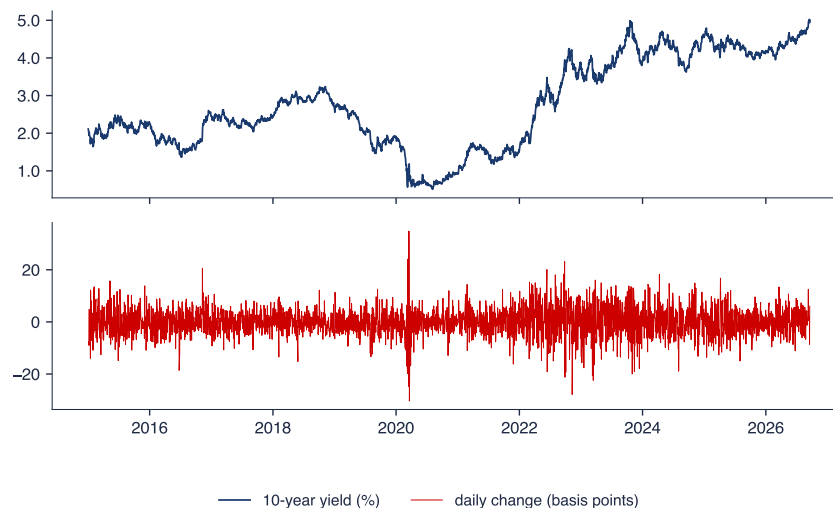


Figure 1: The 10-year US Treasury yield and its daily change, 2015–2026.

- Compute the Ljung–Box statistic $Q(5) = n(n+2) \sum_{k=1}^5 \hat{\rho}_k^2 / (n-k)$ for the daily change.
- State the decision of the test at the 5% significance level.
- Explain why autocorrelations of the level close to 1 do not indicate a stationary series.

Problem 3 • A white noise that is not independent

Let $Y_t = \varepsilon_t \varepsilon_{t-1}$, where ε_t are independent $N(0, \sigma^2)$ variables, so that $E(\varepsilon_t^4) = 3\sigma^4$.

- Compute the mean and the variance of Y_t .
- Show that $\text{Cov}(Y_t, Y_{t-k}) = 0$ for every $k \geq 1$.
- Compute the correlation between Y_t^2 and Y_{t-1}^2 .

Chapter 2. ARMA models

Problem 4 • The AR(2) model of the sunspots

The yearly sunspot number (1700–2008) was modelled with an AR(2) with constant, $y_t - \mu = \phi_1(y_{t-1} - \mu) + \phi_2(y_{t-2} - \mu) + \varepsilon_t$ (in statsmodels, const is the mean μ).

Software output
 ARIMA(2,0,0) with constant, sunspots 1700–2008 (n = 309)
 Log likelihood -1307.32 AIC 2622.64 BIC 2637.57

	coef	std err	z	P> z
const	49.7462	3.9385	12.631	0.000
ar.L1	1.3906	0.0369	37.694	0.000
ar.L2	-0.6886	0.0356	-19.363	0.000
sigma2	274.7272	18.8973	14.538	0.000

The observed values are $y_{2007} = 7.5$ and $y_{2008} = 2.9$.

- Check the stationarity conditions of the estimated AR(2) model.
- Show that the roots of the autoregressive polynomial are complex.
- Compute the period of the cycle from $\cos(2\pi/P) = \phi_1 / (2\sqrt{-\phi_2})$.
- Compute the forecast for 2009.

Problem 5 • Choosing the ARMA order: US industrial production

For the monthly growth of US industrial production, $g_t = 100 \Delta \ln \text{INDPRO}_t$, six ARMA models with constant were estimated.

Software output

Model selection, US industrial production growth (100 dlog INDPRO)
monthly, 1990–02 to 2025–12, n = 431

model	AIC	BIC	LB(12) p
ARMA(1,0)	1235.43	1247.62	0.001
ARMA(2,0)	1226.92	1243.19	0.034
ARMA(0,1)	1229.32	1241.52	0.009
ARMA(0,2)	1227.98	1244.25	0.029
ARMA(1,1)	1229.09	1245.36	0.017
ARMA(3,0)	1228.58	1248.91	0.027

The estimated AR(1) model is $g_t = 0.116 + 0.202(g_{t-1} - 0.116) + \varepsilon_t$, with $\hat{\sigma}^2 = 1.015$; in December 2025, $g_T = 0.454$.

- State the model chosen by the AIC and the model chosen by the BIC.
- Interpret the column of Ljung–Box p -values of the residuals.
- Compute the AR(1) forecasts for January and February 2026.
- Compute the standard deviation of the two-step forecast error.

Problem 6 • The autocorrelation of an MA(1)

Let $y_t = \varepsilon_t + \theta\varepsilon_{t-1}$.

- Show that $\rho(1) = \theta/(1 + \theta^2)$ and $\rho(k) = 0$ for $k \geq 2$.
- Find the values of θ for which $\rho(1) = 0.4$.
- Justify the choice of one of the two values.

Chapter 3. Unit roots and ARIMA models

Problem 7 • The ADF test for the gold price

The augmented Dickey–Fuller (ADF) test was applied to the log of the monthly gold price (USD per ounce, January 2010 – August 2026), in levels and in first differences.

Software output

(1) log gold price, monthly, 2010–01 to 2026–08
Augmented Dickey–Fuller test, deterministic terms: constant and trend

ADF statistic	-0.4071	p-value	0.9867
lags (AIC)	0	observations	199
critical values:	1% -4.005	5% -3.433	10% -3.140

(2) first difference of the log price
Augmented Dickey–Fuller test, deterministic terms: constant

ADF statistic	-14.0405	p-value	0.0000
lags (AIC)	0	observations	198
critical values:	1% -3.464	5% -2.876	10% -2.575

- State the decision of the test for the level at the 5% significance level.
- State the decision of the test for the first difference.
- State the order of integration of the series.
- Explain the choice of the deterministic terms in the two regressions.

Problem 8 • An ARIMA(0,1,1) forecast with drift: gold

For $y_t = 100 \ln P_t$ (the monthly gold price) the model $\Delta y_t = c + \varepsilon_t + \theta\varepsilon_{t-1}$ was estimated.

Software output

ARIMA(0,1,1) with drift, $y = 100 \log(\text{gold})$, monthly, 2010–01 to 2026–08

	coef	std err	z	P> z
drift	0.7105	0.3285	2.163	0.031
ma.L1	-0.0118	0.0639	-0.185	0.853
sigma2	21.7817	2.0883	10.431	0.000

Last observation (2026–08): $y = 840.117$ residual = 8.731

- Test the significance of θ .
- State the simpler model suggested by the result.
- Compute the forecasts for September and October 2026 with the estimated model.
- Compute the standard deviations of the one-step and two-step forecast errors, given that $\psi_1 = 1 + \theta$.
- Convert the forecast for September 2026 into a price (USD per ounce).

Problem 9 • ADF and KPSS for US consumer prices

The ADF test (null hypothesis: unit root) and the KPSS test (null hypothesis: stationarity) were applied to the US consumer price index (CPIAUCSL, monthly, 1990–2025).

Software output					
US CPI (CPIAUCSL), monthly, 1990–2025; inflation = 1200 dlog(CPI), % per year					
series	det.	ADF	ADF p	KPSS	KPSS p
log CPI	ct	-1.985	0.610	0.260	0.010
inflation	c	-3.996	0.001	0.203	0.100
d(inflation)	c	-9.371	0.000	0.099	0.100
KPSS 5% critical values: 0.463 (constant), 0.146 (constant and trend)					
KPSS p-values are truncated to the interval [0.01, 0.10]					

- State the joint conclusion of the two tests for ln CPI.
- State the joint conclusion for the inflation rate.
- State the order of integration of ln CPI.

Chapter 4. Seasonality and forecasting

Problem 10 • The airline model for the CO₂ concentration

The monthly CO₂ concentration at Mauna Loa (ppm, 1960–2000) was modelled with a SARIMA(0, 1, 1)(0, 1, 1)₁₂.

Software output				
SARIMAX(0,1,1)x(0,1,1,12), CO2 at Mauna Loa (ppm), monthly, 1960–01 to 2000–12				
No. observations	492	Log likelihood	-101.50	AIC 209.00
	coef	std err	z	P> z
ma.L1	-0.3522	0.0418	-8.432	0.000
ma.S.L12	-0.8583	0.0286	-30.008	0.000
sigma2	0.0865	0.0056	15.388	0.000
Ljung-Box Q(24) of the residuals: 19.42 p-value 0.619				

- Write the equation of the model with the lag operator.
- Compute the coefficient of ε_{t-13} .
- State the decision of the Ljung–Box test for the residuals ($24 - 2 = 22$ degrees of freedom; $\chi^2_{0.95}(22) = 33.92$).
- Compute the theoretical autocorrelations $\rho(1)$ and $\rho(12)$ of the series $(1 - L)(1 - L^{12})y_t$.

Problem 11 • Forecast evaluation: tourist nights in Romania

For the monthly number of nights spent in tourist accommodation in Romania (millions, Eurostat), the year 2025 was forecast from December 2024 with the seasonal naive method and with a SARIMA model (Figure 2).

Software output			
Nights spent in tourist accommodation, Romania (millions), test year 2025			
month	actual	s.naive	SARIMA
2025-01	1.624	1.441	1.566
2025-02	1.575	1.630	1.538
2025-03	1.574	1.601	1.454
2025-04	1.716	1.698	1.256
2025-05	2.256	2.116	1.595
2025-06	2.836	2.830	2.464
2025-07	4.467	4.660	4.276

2025-08	5.117	5.346	5.000
2025-09	2.710	2.778	2.810
2025-10	2.262	2.323	2.068
2025-11	1.840	1.948	1.676
2025-12	1.707	1.820	1.505
MAE		0.100	0.223

In-sample MAE of the seasonal naive method (2023–2024): 0.162

SARIMA = log-SARIMA(1,0,0)(0,1,1)12, 2015–2024, pandemic dummy 2020-03..2021-05

For the differences $d_t = |e_t^{\text{naive}}| - |e_t^{\text{SARIMA}}|$ of the 12 months, the mean is $\bar{d} = -0.1234$ and the standard deviation is $s_d = 0.210$.

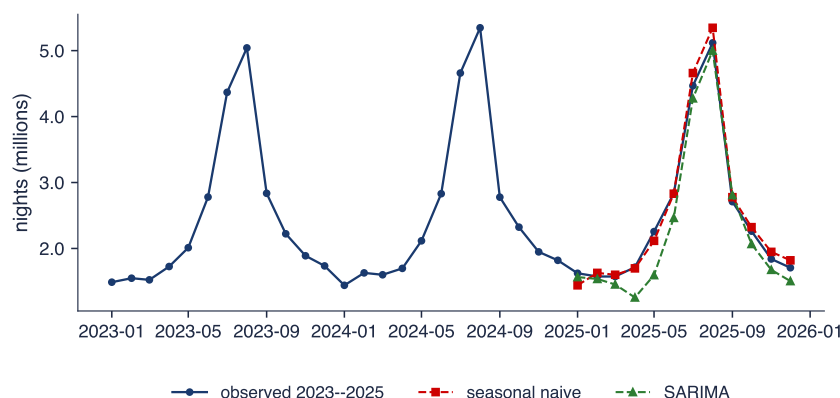


Figure 2: Monthly nights in Romania and the forecasts for 2025.

- Compute the MASE of both methods.
- Compute the Diebold–Mariano statistic $DM = \bar{d}/(s_d/\sqrt{12})$.
- State the decision of the Diebold–Mariano test at the 5% significance level.
- Explain the weak result of the SARIMA model.

Problem 12 • The seasonal difference

In July and August 2024 there were 4.660 and 5.346 million nights, and in July and August 2025, 4.467 and 5.117 million.

- Compute $\Delta_{12} \ln y_t = \ln y_t - \ln y_{t-12}$ for July and August 2025, in %.
- Show that $1 - L^{12} = (1 - L)(1 + L + \dots + L^{11})$.
- Show that Δ_{12} removes a deterministic seasonal component $s_t = s_{t-12}$.

Chapter 5. Conditional volatility and GARCH models

Problem 13 • GARCH(1,1)-t for the DAX

For the daily log returns of the DAX index the model $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ with Student- t innovations was estimated.

Software output				
Constant mean – GARCH(1,1), Student-t innovations, DAX daily log returns (%)				
2010-01-05 to 2026-09-18, n = 4244 Log likelihood -6203.10				
	coef	std err	t	P> t
mu	0.0859	0.0136	6.321	0.000
omega	0.0364	0.0107	3.387	0.001
alpha[1]	0.1170	0.0200	5.855	0.000
beta[1]	0.8645	0.0230	37.630	0.000
nu	5.4683	0.4567	11.973	0.000

- Compute the persistence of volatility.
- Compute the long-run volatility, daily and annualised (252 days).

- (c) Compute the half-life of a volatility shock.
- (d) Compute the excess kurtosis of the Student- t innovations, $6/(\nu - 4)$.
- (e) Interpret the excess kurtosis of the innovations.

Problem 14 • One GARCH step and VaR 1%: Ethereum

A GARCH(1,1) with Normal innovations, estimated on the daily returns of Ethereum (2018–2026), has $\hat{\mu} = 0.083$; $\hat{\omega} = 0.503$; $\hat{\alpha} = 0.064$; $\hat{\beta} = 0.911$. On 18 September 2026 the conditional variance was $\sigma_T^2 = 10.225$ and the return was $r_T = 6.496\%$. Use $z_{0.01} = -2.326$.

- (a) Compute the conditional variance for 19 September 2026.
- (b) Compute the VaR 1% for 19 September 2026, in %.
- (c) Compute the corresponding loss for a position of USD 10,000.

Problem 15 • The asymmetry of volatility: GJR-GARCH for the DAX

On the same DAX data a GJR-GARCH(1,1,1) with Student- t innovations was estimated, $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, where $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$.

Software output				
Constant mean – GJR-GARCH(1,1,1), Student-t, DAX daily log returns (%), 2010–2026				
	coef	std err	t	P> t
omega	0.0397	0.0104	3.795	0.000
alpha[1]	0.0000	0.0096	0.000	1.000
gamma[1]	0.2088	0.0335	6.227	0.000
beta[1]	0.8687	0.0225	38.648	0.000
Log likelihood	-6134.20	(GARCH(1,1)-t: -6203.10)		

- (a) Test the GARCH(1,1)- t model against the GJR-GARCH with a likelihood ratio test.
- (b) Compute σ_t^2 after a shock $\varepsilon_{t-1} = -2$ and after a shock $\varepsilon_{t-1} = +2$, with $\sigma_{t-1}^2 = 1$.
- (c) Compute the persistence $\alpha + \gamma/2 + \beta$.
- (d) Interpret the estimate $\hat{\alpha} = 0$.

Chapter 6. VAR models and Granger causality

Problem 16 • A macroeconomic VAR: United States, 1959–2009

For the vector of real GDP growth (annualised, %), inflation and the 3-month Treasury bill rate (quarterly data, the macrodata set) the information criteria and the Granger causality tests in a VAR(2) with constant were computed.

Software output			
VAR lag order selection, y = (gdp growth, inflation, T-bill rate), US 1959Q2–2009Q3			
p	AIC	BIC	HQIC
0	6.456	6.507	6.477
1	3.661	3.864	3.743
2	3.523	3.877	3.666
3	3.399	3.904	3.603
4	3.411	4.068	3.677
5	3.404	4.212	3.731
6	3.353	4.313	3.742
7	3.438	4.550	3.888
8	3.449	4.712	3.960
chosen: AIC 6, BIC 1, HQIC 3			

Granger causality F-tests in the VAR(2) (H0: the cause does not Granger-cause)			
cause → effect	F	df	p-value
rate → gdp	2.747	(2, 579)	0.0649
gdp → rate	3.118	(2, 579)	0.0450
infl → rate	3.103	(2, 579)	0.0457

- (a) State the order chosen by each information criterion.

- (b) Compute the number of coefficients of a VAR(6) and of a VAR(1) with three variables and a constant.
- (c) Interpret the Granger causality test from the interest rate to GDP growth.
- (d) Interpret the Granger causality test from GDP growth to the interest rate.

Problem 17 • The stability and the forecast of a VAR(1)

A VAR(1) for $x_t = (g_t, \Delta i_t)'$, US real GDP growth and the change in the 3-month interest rate, has the equations

$$g_t = 2.259 + 0.273 g_{t-1} + 0.612 \Delta i_{t-1} + u_{1t}, \quad \Delta i_t = -0.115 + 0.031 g_{t-1} + 0.006 \Delta i_{t-1} + u_{2t}.$$

In the third quarter of 2009, $g_T = 2.74$ and $\Delta i_T = -0.06$.

- (a) Compute the eigenvalues of the coefficient matrix from its trace and determinant.
- (b) Compute the forecast for the fourth quarter of 2009.
- (c) Compute the unconditional mean of the process, $\mu = (I - A)^{-1}c$.

Chapter 7. Cointegration and VECM

Problem 18 • The Engle–Granger method: US 10-year and 3-month rates

The Engle–Granger method was applied to the 10-year (GS10) and 3-month (TB3MS) US Treasury yields, monthly data 1990–2025. The 5% Engle–Granger critical value for two variables with a constant is -3.34 .

Software output				
Step 1: OLS of GS10 on TB3MS, US monthly 1990–01 to 2025–12 (% per year)				
	coef	std err	t	P> t
const	2.2746	0.0824	27.593	0.000
TB3MS	0.7286	0.0236	30.903	0.000
R-squared 0.690 Durbin-Watson 0.038 n = 432				

Step 2: ADF test on the residuals u_t (no constant, lags by AIC)
 ADF statistic -2.5599 lags 3 (Dickey–Fuller table p-value 0.0101)

- (a) Interpret the comparison between the R^2 and the Durbin–Watson statistic.
- (b) State the decision of the cointegration test at the 5% significance level.
- (c) Explain why the p -value from the Dickey–Fuller table cannot be used in step 2.

Problem 19 • The Johansen test and the VECM: the term structure

On the longer sample 1960–2025 the Johansen tests were applied and a VECM with rank 1 was estimated, $\Delta x_t = \alpha \beta' x_{t-1} + \dots$, with $x_t = (\text{GS10}_t, \text{TB3MS}_t)'$.

Software output				
Johansen test, (GS10, TB3MS), US monthly 1960–01 to 2025–12, constant, 2 lagged differences				
H0: rank <=	trace	5% c.v.	max-eig	5% c.v.
0	26.25	15.49	23.43	14.26
1	2.83	3.84	2.83	3.84
VECM, rank 1, constant in the cointegrating relation				
beta (normalised): GS10 1.000 TB3MS -1.026 const -1.276				
alpha	coef	std err	z	
D(GS10)	-0.0174	0.0072	-2.41	
D(TB3MS)	0.0236	0.0106	2.23	

In December 2025, $\text{GS10} = 4.14$ and $\text{TB3MS} = 3.59$.

- (a) Determine the cointegration rank with the trace statistic.
- (b) Write the long-run equilibrium relation.
- (c) Interpret the long-run equilibrium relation.
- (d) Compute the half-life of a deviation from equilibrium, with $z_t = z_{t-1}(1 + \alpha_1 + \beta_2 \alpha_2)$.
- (e) Compute the deviation from equilibrium in December 2025.

- (f) Compute the expected corrections of the two rates.

Chapter 8. Long memory and ARFIMA models

Problem 20 • The GPH estimator for the VIX index

The memory parameter d of the log of the VIX index (daily, 2004–2026) was estimated with the Geweke–Porter–Hudak (GPH) log-periodogram regression.

Software output					
GPH log-periodogram regression, bandwidth $m = n^{0.5}$					
series	n	m	d_hat	s.e.	asy s.e.
log VIX	5717	75	0.634	0.086	0.074
d log VIX	5716	75	-0.376	0.087	0.074
asy s.e. = $\pi / \sqrt{24 m}$; daily data 2004-01-02 to 2026-09-18					

- Compute the 95% confidence interval for d with the asymptotic standard error.
- Test the hypothesis $d = 0.5$.
- Interpret the result of the test.
- Check the consistency of the estimate for the series with the estimate for its first difference.
- Compute the first three weights π_k of $(1 - L)^d$ for $d = 0.63$, with $\pi_k = \pi_{k-1}(k - 1 - d)/k$.

Problem 21 • The long memory of volatility: gold

Figure 3 shows the autocorrelation function of the daily absolute returns of gold (2010–2026, $n = 4350$). The sample values are $\hat{\rho}(1) = 0.110$; $\hat{\rho}(10) = 0.164$; $\hat{\rho}(100) = 0.081$.

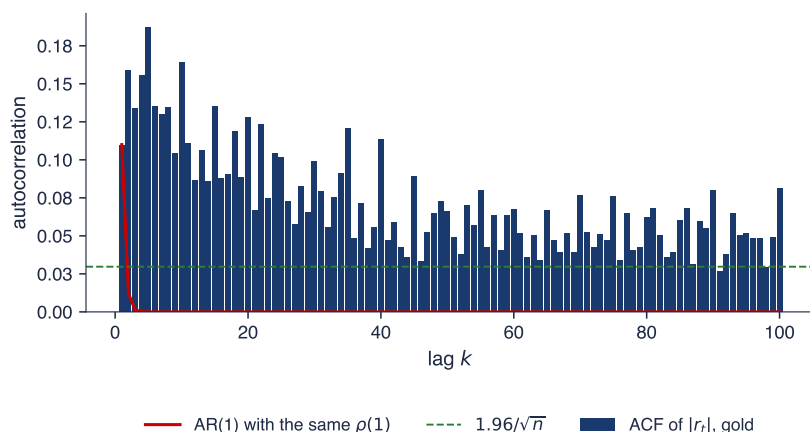


Figure 3: Autocorrelations of $|r_t|$ for gold and of an AR(1) with the same $\rho(1)$.

- Compute $\rho(10)$ and $\rho(100)$ for an AR(1) with $\rho(1) = 0.110$.
- Estimate d from the slope of the power law $\rho(k) \approx Ck^{2d-1}$, using lags 10 and 100.
- Name a suitable model for this volatility.

Chapter 9. Machine learning for time series

Problem 22 • The sign of the Bitcoin return

Two classifiers forecast the sign of the next-day Bitcoin return, with five lags and the 20-day volatility as features.

Software output	
Direction of the next-day Bitcoin return (1 = up), features: 5 lags, 20-day vol	
train 2018-01 to 2023-12 ($n = 2170$), test 2024-01-01 to 2026-09-18 ($n = 992$)	
model	test accuracy
logistic regression	0.4980

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random forest (300 trees)          0.4950
always 'up' (majority class)      0.5040
share of up days in the training set: 0.5111
```

- Compute the statistic $z = (\hat{a} - a_0) / \sqrt{a_0(1 - a_0)/n}$ for the random forest against the “always up” benchmark.
- Explain why the right benchmark is the majority class and not an accuracy of 50%.
- Explain why cross-validation with random folds would overstate the accuracy.

Problem 23 • Conformal intervals and the pinball loss: gold

For the daily return of gold the point forecast is 0, and the conformity scores are the absolute errors. The 20 calibration scores (sorted) are: 0.13; 0.14; 0.32; 0.53; 0.63; 0.68; 0.77; 1.00; 1.14; 1.15; 1.38; 1.55; 1.66; 1.79; 1.81; 2.12; 2.14; 2.90; 3.01; 4.05. The returns of 14–18 September 2026 are -1.39 ; -0.12 ; -0.27 ; 1.73 ; 0.77 .

- Compute the 90% conformal interval, with the score of order $\lceil (n+1)(1-\alpha) \rceil$.
- Compute the share of the returns of 14–18 September 2026 that fall inside the interval.
- Compute the mean pinball loss, with $\tau = 0.05$, of the quantile forecasts $q_A = -1.2$ and $q_B = -2.5$.

Chapter 10. State space models, the Kalman filter and Markov switching

Problem 24 • The Kalman filter for the US unemployment rate

The monthly US unemployment rate (2022–2025) was modelled with a local level model, $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$. The value for October 2025 is missing (it was not published).

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Software output
-----
UnobservedComponents, local level, US unemployment rate (UNRATE, %),
monthly 2022-01 to 2025-12 (2025-10 missing), n = 48   Log likelihood 36.07
coef      std err      z      P>|z|
-----
sigma2.irregular    0.0037    0.0020    1.795    0.073
sigma2.level        0.0059    0.0030    1.973    0.048
filtered level 2025-12: 4.4213   variance P(t|t): 0.0026
```

In January 2026 the unemployment rate was 4.3%.

- Compute the signal-to-noise ratio $q = \sigma_\eta^2 / \sigma_\varepsilon^2$.
- Compute the prediction variance, the Kalman gain and the filtered level for January 2026.
- Describe the filter step for the missing month (October 2025).

Problem 25 • Volatility regimes: Nasdaq 100

The weekly returns of the Nasdaq 100 index (2015–2026) were modelled with a two-regime Markov switching model.

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Software output
-----
MarkovRegression, 2 regimes, switching mean and variance,
Nasdaq 100 weekly log returns (%), 2015-01-09 to 2026-09-18, n = 611
regime      mean      variance  stay prob.
-----
calm         0.548      3.200      0.9598
turbulent    -0.027     14.335     0.9378
share of weeks with smoothed P(turbulent) > 0.5: 0.376
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- Compute the expected duration of each regime, $1/(1 - p_{ii})$, in weeks.
- Compute the ergodic probability of the turbulent regime.
- Compare the ergodic probability with the observed share of turbulent weeks.
- Compute the annualised volatility (52 weeks) of each regime.

Answers

- (a) $\pm 1.96/\sqrt{100} = \pm 0.196$. (b) $Q(10) = 88.13 > 18.31$: white noise is rejected. (c) AR(1): the ACF decays gradually (8 of the first 10 autocorrelations are outside the band) and the PACF is significant only at lag 1 (0.498).
- (a) $Q(5) = 8.52$. (b) $8.52 < 11.07$: the hypothesis of uncorrelated changes is not rejected (although $\hat{\rho}_1 = -0.037$ is at the edge of the band ± 0.036). (c) For a random walk the sample ACF stays close to 1 over many lags; the level does not revert to a constant mean, and differencing makes it stationary.
- (a) $E(Y_t) = 0$; $\text{Var}(Y_t) = \sigma^4$. (b) Every term of the product contains an ε that appears only once, with mean 0. (c) $\text{Cov}(Y_t^2, Y_{t-1}^2) = 3\sigma^8 - \sigma^8 = 2\sigma^8$; $\text{Var}(Y_t^2) = 9\sigma^8 - \sigma^8 = 8\sigma^8$; the correlation is 0.25: white noise, but not independent.
- (a) $\phi_1 + \phi_2 = 0.702 < 1$; $\phi_2 - \phi_1 = -2.079 < 1$; $|\phi_2| = 0.689 < 1$: stationary. (b) $\phi_1^2 + 4\phi_2 = -0.821 < 0$. (c) $\cos(2\pi/P) = 0.838$; $P = 10.9$ years. (d) $\hat{y}_{2009} = 13.69$.
- (a) AIC: ARMA(2,0); BIC: ARMA(0,1). (b) All p -values are below 0.05: autocorrelation remains in the residuals (least for the AR(2), $p = 0.034$); the models are too simple. (c) 0.184; 0.130. (d) $\sqrt{1.015(1 + 0.202^2)} = 1.028$.
- (a) $\gamma(0) = \sigma^2(1 + \theta^2)$, $\gamma(1) = \theta\sigma^2$, $\gamma(k) = 0$ for $k \geq 2$. (b) $0.4\theta^2 - \theta + 0.4 = 0$: $\theta = 0.5$ or $\theta = 2$. (c) $\theta = 0.5$, the only invertible value ($|\theta| < 1$).
- (a) $-0.407 > -3.433$: the unit root is not rejected. (b) $-14.04 < -2.876$: rejected. (c) $I(1)$. (d) The level trends, hence constant and trend; the difference fluctuates around a non-zero mean, hence a constant.
- (a) $z = -0.185$, $p = 0.853$: θ is not significant. (b) A random walk with drift is enough. (c) 840.723; 841.434. (d) 4.667; 6.561. (e) $e^{8.40723} = 4479.3$ USD per ounce.
- (a) ADF does not reject the unit root ($p = 0.610$), KPSS rejects stationarity ($0.260 > 0.146$): unit root. (b) ADF rejects ($p = 0.001$), KPSS does not reject ($0.203 < 0.463$): inflation is stationary. (c) $\ln \text{CPI} \sim I(1)$.
- (a) $(1-L)(1-L^{12})y_t = (1-0.352L)(1-0.858L^{12})\varepsilon_t$. (b) The coefficient of ε_{t-13} is $\theta\Theta = 0.302$. (c) $Q(24) = 19.42 < 33.92$ ($p = 0.619$): the residuals are white noise. (d) $\rho(1) = -0.313$; $\rho(12) = -0.494$.
- (a) 0.617 (seasonal naive); 1.381 (SARIMA). (b) $\text{DM} = -2.04$. (c) $|-2.04| > 1.96$: equal accuracy is rejected; the seasonal naive method is better. (d) The model strongly underestimates the spring (April: 1.256 against 1.716; May: 1.595 against 2.256); estimated on 2015–2024, with the pandemic years, it misses the level reached after 2022, which the seasonal naive method takes directly from 2024.
- (a) July: -4.23% ; August: -4.38% . (b) The product telescopes: $(1-L)\sum_{j=0}^{11} L^j = 1 - L^{12}$. (c) $(1-L^{12})s_t = s_t - s_{t-12} = 0$.
- (a) $\alpha + \beta = 0.9815$. (b) $\hat{\sigma}^2 = 0.0364/0.0185 = 1.968$; 1.403% per day; 22.27% per year. (c) $\ln 0.5 / \ln 0.9815 = 37.1$ days. (d) $6/1.468 = 4.09$. (e) Fat tails; the kurtosis is finite because $\nu > 4$.
- (a) $0.503 + 0.064 \cdot 6.413^2 + 0.911 \cdot 10.225 = 12.45$ ($\varepsilon_T = 6.413$). (b) $\sigma = 3.528$; VaR 1% = $2.326 \cdot 3.528 - 0.083 = 8.12\%$. (c) $10,000(1 - e^{-0.0812}) = 780$ USD.
- (a) $\text{LR} = 2(-6134.20 + 6203.10) = 137.8 > 3.84$: the GJR-GARCH is chosen. (b) 1.744 and 0.908. (c) 0.973. (d) Only negative shocks raise the next day's volatility (the leverage effect).
- (a) AIC: 6; BIC: 1; HQIC: 3. (b) $3(1+18) = 57$ and $3(1+3) = 12$. (c) $p = 0.065$: not rejected at 5%, rejected at 10%. (d) $p = 0.045$: GDP growth Granger-causes the interest rate (the monetary policy reaction).
- (a) Trace 0.279, determinant -0.0173 ; $\lambda_{1,2} = 0.331$ and -0.052 , both below 1 in modulus: the VAR is stable. (b) $\hat{g} = 2.970$; $\hat{\Delta i} = -0.030$. (c) $\mu = (3.091; -0.019)$.
- (a) $R^2 = 0.690 > \text{DW} = 0.038$: the sign of a spurious regression (the Granger–Newbold rule). (b) $-2.56 > -3.34$: the null hypothesis of no cointegration is not rejected. (c) The residuals come from an estimated regression (OLS minimises their variance), so the statistic is shifted towards negative values; the Engle–Granger critical values are needed.
- (a) $26.25 > 15.49$ rejects $r = 0$; $2.83 < 3.84$ does not reject $r \leq 1$: rank 1. (b) $\text{GS10} = 1.276 + 1.026 \text{ TB3MS}$. (c) The 10-year rate follows the 3-month rate almost one for one, plus a term premium of about 1.3 percentage points. (d) $1 - 0.0174 - 1.026 \cdot 0.0236 = 0.958$; 16.3 months. (e) $z = -0.819$. (f) Corrections: $+0.014$ (GS10) and -0.019 (TB3MS) percentage points per month.
- (a) $[0.489; 0.779]$. (b) $z = 1.81 < 1.96$: $d = 0.5$ is not rejected. (c) The series is at the border of stationarity, with long memory and mean reversion ($d < 1$). (d) $-0.376 \approx 0.634 - 1$. (e) -0.630 ; -0.117 ; -0.053 .
- (a) $0.11^{10} = 2.6 \cdot 10^{-10}$; $0.11^{100} \approx 10^{-96}$. (b) Slope $\ln(0.081/0.164)/\ln 10 = -0.306$; $d = 0.35$. (c) A long-memory volatility model: FIGARCH, HAR or ARFIMA for realised volatility.
- (a) $z = (0.495 - 0.504)/\sqrt{0.504 \cdot 0.496/992} = -0.57$. (b) The share of up days is above 50%; a model without information already reaches 50.4%. (c) Targets and neighbouring features overlap in time (the 20-day volatility); random folds put days after the test date into training (leakage).
- (a) $[21 \cdot 0.9] = 19$; $\hat{q} = 3.01$; the interval $[-3.01; 3.01]$. (b) $5/5$. (c) $L_A = 0.105$; $L_B = 0.132$: the quantile q_A has the smaller loss.
- (a) $q = 0.0059/0.0037 = 1.6$. (b) $P = 0.0026 + 0.0059 = 0.0085$; $K = 0.0085/(0.0085 + 0.0037) = 0.697$; $a = 4.421 + 0.697(4.3 - 4.421) = 4.337$. (c) The update is skipped ($a_{t|t} = a_t$, $P_{t|t} = P_t$) and only the prediction is made: the variance grows by σ_η^2 .
- (a) $1/0.040 = 25.0$ weeks (calm); $1/0.062 = 16.1$ weeks (turbulent). (b) $0.040/0.102 = 0.392$. (c) Close to the observed share, 0.376. (d) $\sqrt{52 \cdot 3.2} = 12.9\%$; $\sqrt{52 \cdot 14.335} = 27.3\%$.